

TOWARDS DYNAMIC LCA: A MODEL FOR GLOBAL WARMING IMPACT ASSESSMENT AND ITS APPLICATION TO GHG-EMISSION MITIGATION

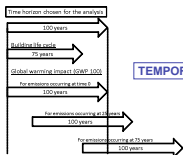
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INTRODUCTION

Current LCA = no temporal resolution

$$\text{Impact score for global warming} = \sum \text{Emissions} \times \text{GWP}_{\text{TH}}$$

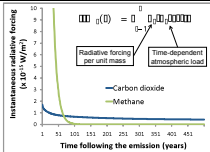


Levasseur et al. 2010

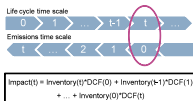
TEMPORAL INCONSISTENCY

MATERIALS AND METHODS

Development of dynamic characterization factors



Dynamic impact on global warming



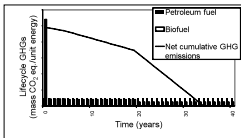
CONCLUSIONS

- Development of dynamic characterization factors for every GHG with variable time horizons to account for the timing of the emissions;
- Applicable to any temporal profile and to any different type of LCA;
- Provides temporal information to the practitioner to help in decision making;
- Dynamic LCA is applicable to any impact category if sets of dynamic characterization factors are developed.

Levasseur, A., Lesage, P., Margni, M., Deschênes, L., Samson, R. 2010. Considering time in LCA: dynamic LCA and its application to global warming impact assessments. Environmental Science & Technology 44(8), pp. 3168-3174.

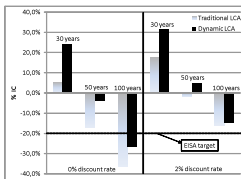
RESULTS & DISCUSSION

Application to the US EPA LCA on renewable fuels



Dynamic inventory

Life cycle GHGs of corn ethanol (including LUC emissions) compared to life cycle GHGs of gasoline.



Levasseur et al. 2010

% Impact change (%IC)

Reduction (negative value) or increase (positive value) in global warming impact for corn ethanol compared to gasoline for different time horizons.

Dynamic LCA

Impact of
corn ethanol relative to
gasoline

Traditional LCA

Impact of
corn ethanol relative to
gasoline

WHY ?

➢ The later the emission occurs, the shorter the time horizon used for assessing its impact should be.

➢ As we approach the chosen time horizon, the credit in GHG emission for ethanol is assessed on a decreasing time period, so that its beneficial impact, which helps to compensate for the LUC emissions, decreases.

ACKNOWLEDGEMENTS

